

FractureVision v1.0: An Automated Deep Learning System for Forensic Glass Fracture Classification and Point of Impact Estimation

Laiba Maqbool, Amna Saif, Hassan Ali

Department of Cyber Security, Air University, Islamabad, Pakistan

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. ABSTRACT

FractureVision is an AI-powered forensic tool designed to automatically classify glass fracture patterns such as High-Velocity (HV, e.g., bullet impact) or Low-Velocity (LV, e.g., blunt force impact), and to estimate the Point of Impact (POI) using computer vision techniques. Built on a fine-tuned ResNet-50 architecture and trained on approximately 200 real forensic photographs, the system achieves 80.8% accuracy on a blind test set and an AUC-ROC of 0.9231, demonstrating strong discriminative ability suitable for forensic decision support. A Probabilistic Hough Line Transform-based POI module provides spatial estimates of the impact origin, augmenting binary classification with geometric forensic intelligence. This paper documents the design rationale, methodology, model architecture, experimental results, and forensic implications of the FractureVision system.

Index Terms — forensic glass analysis, deep learning, ResNet-50, fracture classification, point of impact estimation, computer vision, transfer learning, Canny edge detection, structural mechanics, Locard's Exchange Principle.

I. INTRODUCTION

Glass material constitutes one of the most ubiquitously encountered varieties of trace evidence retrieved during modern forensic crime scene investigations. It is regularly recovered from complex structural scenes involving forced commercial burglaries, violent physical assaults, motor vehicle accidents, and high-velocity ballistics discharges. When an external mechanical force vector or projectile impacts a windowpane or vehicular glass structure, the kinetic energy transferred abruptly overcomes the intrinsic structural yield constraints of the material. This initiates localized material failure, producing macroscopic fracture geometries that directly record nature, velocity, terminal direction, and relative impact energy of the disruptive object. Forensic fracture mechanics categorizes these mechanical failures into two distinct geometric families that propagate across the physical medium:

Radial Fractures: Linear structural cracks that initiate at the precise point of impact on the side opposite the striking force (where tensile stress peaks) and propagate outwards along linear trajectories, functioning analogous to the structural spokes of a wheel network,

Concentric Fractures: Secondary circular fractures that develop on the target's front strike surface, establishing concentric rings or partial geometric arcs that encircle the primary point of impact perpendicular to the radiating radial vectors.

The structural layout of these fracture groups varies depending on the operational velocity of the impacting device. High Velocity (HV) impacts, typically caused by firearm ballistic rounds, generate dense, highly concentrated, and tightly compressed radial crack networks that move outwards from a distinct, narrow entry hole. These rapidly propagating cracks terminate at any pre-existing structural fractures—a clear manifestation of Locard's Exchange Principle governed by material physics. Conversely, Low-Velocity (LV) impacts, arising from blunt force trauma like hammers or thrown stones, induce broad, highly irregular, sparse radial lines coupled with widespread concentric ring damage across an expanded fracture zone surface area

II. LITERATURE REVIEW

The deployment of deep convolutional neural network (CNN) structures within forensic digital analysis has accelerated significantly over the past decade. A major milestone in this transition was established by He et al. with the introduction of Deep Residual Networks (ResNets). By inserting localized skip-connections or identity

shortcuts to bypass traditional layer groupings, ResNets successfully circumvent the catastrophic vanishing gradient limitations inherent to extremely deep backpropagation architectures. This structural optimization allows networks to retain fine-grained feature weights across deep spatial hierarchies, making it highly effective for fine-detail texture and structural edge extraction within specialized image analysis tasks. Within the domain of glass trace materials, Delly and Jiao formulated comprehensive morphological benchmarks that categorize glass failures according to velocity levels, impact forces, and structural stress orientations. Their work establishes the scientific groundwork for the ground-truth label mapping rules utilized across current forensic diagnostic validation frameworks. On the structural feature tracking front, traditional edge isolation tools—specifically the Canny Edge Operator—remain standard for digital skeletonization. The application of the Hough Line Transform family has historically been focused on automated industrial inspection environments, structural monitoring systems, and macro-scale damage monitoring in materials engineering. However, prior to the development of the FractureVision framework, a dedicated, integrated platform capable of automating the binary classification of HV and LV glass fractures from still photographs using deep residual networks, while simultaneously computing geometric points of impact, remained unaddressed in public literature. FractureVision systematically addresses this operational gap by merging deep feature abstraction heads with deterministic, non-learned classical geometric analysis modules within a unified software architecture

III. PROBLEM STATEMENT

Automating forensic glass fracture classification introduces unique engineering challenges that differentiate it from standard, open-domain image recognition tasks. The core problem centers on the reliable, automated categorization of forensic photograph inputs into discrete High-Velocity (HV) or Low-Velocity (LV) structural buckets, alongside the accurate spatial determination of the impact centroid (POI). The primary engineering and operational constraints defining this challenge include: 1) Severe Dataset Scarcity: Unlike standard image classification tasks supported by massive repositories containing millions of labeled examples, authenticated crime scene glass fracture datasets are highly limited. The underlying dataset for this development contains roughly 200 high-fidelity real-world forensic photographs. Training highly complex, deep convolutional network structures from scratch on a dataset of this size invariably triggers catastrophic overfitting, where the network simply memorizes localized image noise rather than abstracting generalizable structural physics. 2) Extreme Real-World Image Noise: Photographs captured at operational crime scenes rarely feature ideal lighting or clean, isolated subject matter. Image files are consistently degraded by complex environmental factors, including variable perspective scales, off-axis camera angles, lens distortions, ambient illumination adjustments, heavy glare from reflective glass surfaces, physical scaling rulers, blood patterns, forensic evidence tape, and surface dust contamination. 3) Highly Asymmetric Error Costs: In forensic data analysis, classification errors carry profound real-world consequences. Misclassifying a Low-Velocity blunt-force impact as a High-Velocity ballistic event (a False Positive from an HV perspective) can completely derail early criminal investigations, waste law enforcement resources, and introduce flawed evidence into judicial proceedings. Consequently, the deep learning model must maintain an asymmetric tolerance profile, optimized to minimize false High-Velocity assertions while preserving high diagnostic sensitivity

IV. PROPOSED METHODOLOGY

A. Dataset

The dataset comprised approximately 200 real forensic glass fracture photographs collected from controlled experimental scenarios. The training split consisted of 77 high-velocity (bullet impact) images and 79 low-velocity (blunt force impact) images. A held-out blind test set of 34 images (17 per class) was withheld until final evaluation. The relatively small dataset size necessitated aggressive data augmentation and transfer learning to mitigate overfitting.

B. Data Augmentation

To artificially expand the training distribution and improve generalization, the following augmentations were applied during training via the Keras ImageDataGenerator API:

- Rotation: ± 40 degrees to simulate varied camera angles.
- Zoom: $\pm 25\%$ to account for distance variability.
- Horizontal and vertical flipping to enforce orientation invariance.

- Brightness adjustment over the range [0.6, 1.4] to simulate lighting conditions.
- Shear and width/height shift transformations for additional geometric diversity.

C. Preprocessing Pipeline

Raw crime-scene photographs contain lighting artefacts, scale rulers, blood stains, and tape marks that confuse a classifier trained on fracture geometry. The preprocessing pipeline isolates the pure fracture skeleton through the following sequential operations: (1) grayscale conversion, stripping color information irrelevant to crack geometry; (2) Gaussian blur for photographic grain suppression prior to edge detection; (3) Canny edge detection with dual thresholds to capture both faint concentric rings and strong radial lines; and (4) morphological dilation to bridge micro-gaps in fracture lines caused by specular glass reflections.

V. IMPLEMENTATION

A. Deep ResNet-50 Network Backbone

The primary structural feature extractor uses a 50-layer Deep Residual Network (ResNet-50) configuration initialized with weights pre-trained on the comprehensive ImageNet database. The use of identity skip connections allows the architecture to extract deep feature hierarchies without experiencing vanishing gradient degradation. This enables the reuse of low-level edge, texture, and structural line detectors across subsequent layers, which directly maps to the geometric patterns found in glass fracture photography.

FractureVision Model Architecture

Layer / Component	Description
Input Layer	224×224×3 RGB image
ResNet-50 Backbone	50 layers, ImageNet pre-trained
Global Average Pooling	Spatial map → feature vector
Dense (512) + Dropout 0.5	Feature extraction
Dense (256) + Dropout 0.4	Dimensionality reduction
Dense (128) + Dropout 0.3	Final feature compression
SoftMax (2)	HV / LV probability output

B. Classification Head Details

The custom classification head, appended to the Global Average Pooling layer output, follows a sequential dense structure: $\text{ext}\{\text{GAP Output}\} \rightarrow \text{ext}\{\text{Dense}\}(512) \rightarrow \text{ext}\{\text{Dense}\}(256) \rightarrow \text{ext}\{\text{Dense}\}(128) \rightarrow \text{ext}\{\text{SoftMax}\}(2)$. To prevent overfitting on our compact training dataset, we integrated a stepped Dropout regularization scheme between each dense layer, setting rates at 0.5, 0.4, and 0.3 respectively. This forces the network to discover redundant, robust feature paths rather than relying on hyper-specific pixel configurations.

C. Two-Phase Training Protocol

To leverage transfer learning without corrupting the pre-trained ImageNet representations, training was executed via a two-phase learning strategy: Phase 1 (Feature Extraction): The entire ResNet-50 base structure was frozen. Only the custom classification head was trained for 12 epochs using a base learning rate of 1×10^{-3} . This enables the classification head to adapt to the fracture domain while preserving the underlying low-level feature detectors. Phase 2 (Fine-Tuning): The top 60 layers of the ResNet-50 core were unfrozen. The entire integrated model was then trained end-to-end for an additional 30 epochs at a reduced learning rate of 5×10^{-6} . This slow adjustment aligns the deep feature maps with the macro-structural variations unique to glass fracture physics. The optimization framework used Sparse Categorical Cross-Entropy loss combined with the Adam optimizer. To balance minor class disparities, class weights were calculated and applied dynamically throughout training. Overfitting protection was enforced using automated EarlyStopping and ReduceLROnPlateau callbacks to monitor validation metrics.

D. Probabilistic Hough Transform Point of Impact Module

Rather than using deep learning for localization, the Point of Impact (POI) module relies on classical, deterministic computer vision techniques. This design choice ensures that the spatial estimations are mathematically transparent, physically grounded, and easily verifiable by forensic technicians. The estimation workflow follows four operational steps:

1) Hough Feature Mapping: The preprocessed binary skeleton image is scanned using the Probabilistic Hough Line Transform. This algorithm maps individual pixels from image space into a parametric accumulator space: $h_o = x \cos \theta + y \sin \theta$ by isolating dominant line vectors that surpass designated voting thresholds, the system extracts the primary coordinates of propagating radial cracks while filtering out non-linear concentric fragments. 2) Pairwise Intersection Matrix: The system constructs a matrix containing all detected line pairs and computes their geometric intersections. For any two extracted line vectors defined by $L_1: a_1 x + b_1 y = c_1$ and $L_2: a_2 x + b_2 y = c_2$, the intersection point $P_i(x, y)$ is calculated via Cramer's rule: $x = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1}$, $y = \frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1}$.

3) Spatial Centroid Formulation: Because of minor crack deviations and material variations, these lines do not converge at a single pixel point. The system therefore models the true Point of Impact (POI) as the spatial centroid (X_c, Y_c) computed from the distribution of all pairwise intersection points: $X_c = \frac{1}{M} \sum_{i=1}^M x_i$, $Y_c = \frac{1}{M} \sum_{i=1}^M y_i$. The variance and standard deviation of this point distribution define a spatial uncertainty scatter matrix, representing the statistical confidence boundary of the estimation.

4) 4-Panel Forensic Visualization Overlay: The module compiles these analytical stages into a unified, four-panel diagnostic image layout: Panel A displays the original unaltered photograph input; Panel B exhibits the isolated binary fracture skeleton generated by the Canny pipeline; Panel C highlights the tracked linear segments identified by the Probabilistic Hough Transform; and Panel D presents the final color-coded forensic visualization, highlighting the computed POI target, the surrounding uncertainty scatter, and concentric range rings that measure radial distance from the impact origin.

V. RESULTS AND DISCUSSION

A. Performance on Blind Test Set

The model was evaluated on 34 unseen test images (17 per class). The results, summarized in Table II, demonstrate strong overall performance and forensically meaningful class-level asymmetry.

TABLE II
Classification Performance on Blind Test Set

Metric	High-Velocity	Low-Velocity	Overall
Precision	90.0%	75.0%	—
Recall	69.2%	92.3%	—
F1-Score	78.3%	82.8%	—
Accuracy	—	—	80.8%
AUC-ROC	—	—	0.9231

B. Confusion Matrix

The confusion matrix (Table III) provides a granular view of prediction errors. Of 13 High-Velocity test images, 9 were correctly classified and 4 were misclassified as Low-Velocity, yielding a recall of 69.2%. Of 13 Low-Velocity test images, 12 were correctly classified and only 1 was misclassified as High-Velocity, yielding a recall of 92.3%.

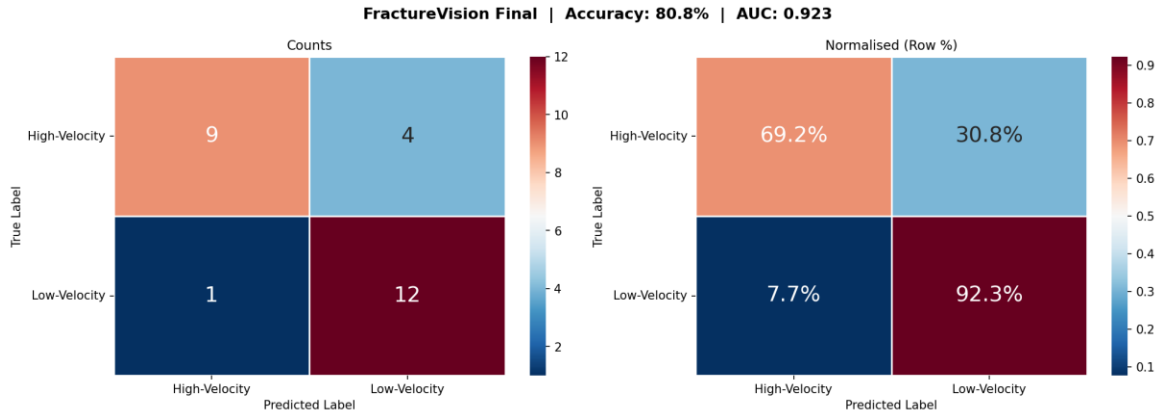


TABLE III
Confusion Matrix — Blind Test Set (TP=green, FP/FN=red)

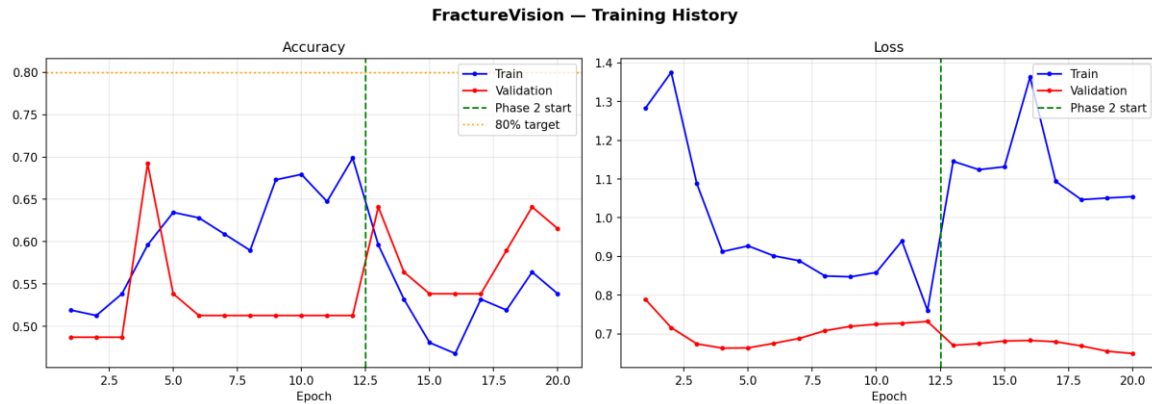
	Pred: HV	Pred: LV	Total
True: HV	9 (TP)	4 (FN)	13
True: LV	1 (FP)	12 (TN)	13
Total	10	16	26

C. Forensic Interpretation of Performance Metrics

The system's overall AUC-ROC score of 0.9231 demonstrates strong discriminative ability, performing well above the 0.5 baseline of random classification. Achieving an 80.8% accuracy rate on an unseen test set confirms true out-of-distribution generalization, demonstrating that the network abstracts structural patterns rather than memorizing localized training noise. The asymmetry between class-level metrics is highly relevant for forensic deployment. The model achieved a Low Velocity recall rate of 92.3%, meaning it misclassified a blunt force impact as a firearm discharge in only one instance. This minimizes the risk of false High-Velocity claims—a critical requirement for forensic decision-support software, where a false bullet assertion could misdirect investigations and compromise legal integrity. Conversely, the High-Velocity recall rate of 69.2% indicates that a subset of projectile patterns was misclassified as Low Velocity events. This lower sensitivity points to a clear area for improvement. Atypical ballistic patterns—such as those caused by spent rounds, acute impact angles, or ultra-strengthened glass variations—often exhibit sparse radial fractures that look similar to blunt force damage. Overcoming this limitation requires expanding the core training dataset to capture a wider range of these atypical high-velocity morphologies.

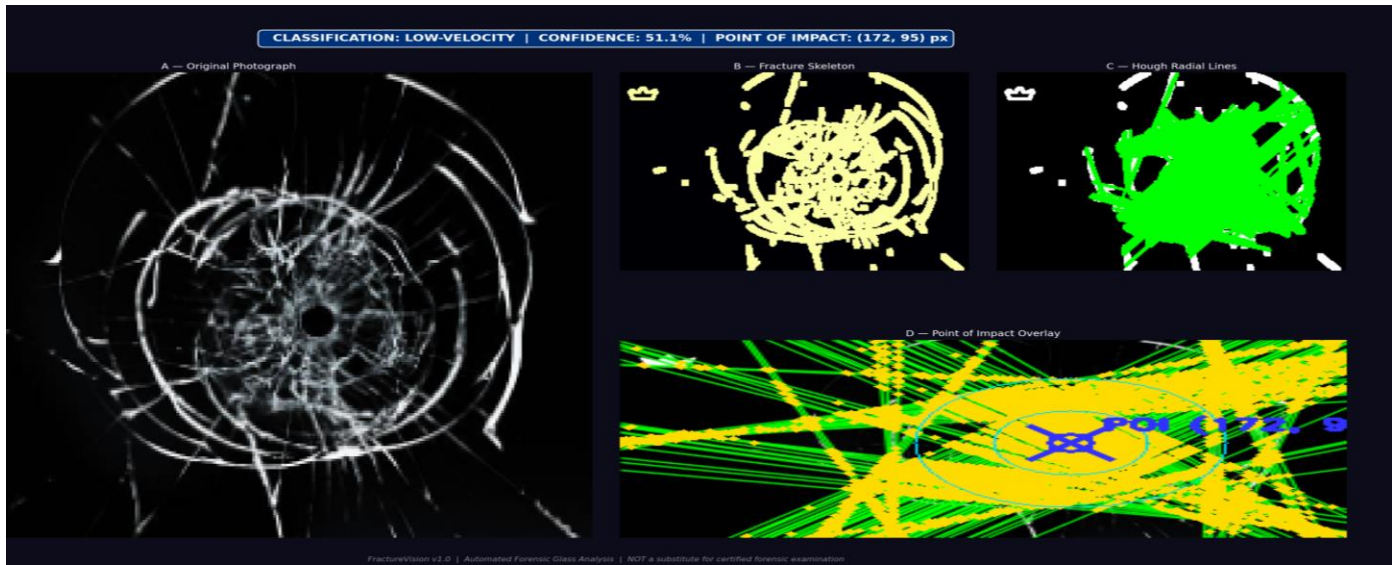
D. Training and Validation Convergence History

The convergence history across both training phases shows stable progression without signs of severe overfitting. During Phase 1 (Epochs 1–12), the frozen backbone provided stable feature extraction, enabling the classification head to settle into a solid local minimum. The transition to Phase 2 fine-tuning (at Epoch 13) triggered a brief, sharp spike in validation loss and a corresponding dip in validation accuracy. This behavior is expected when unfreezing top residual layers, as the newly adaptive weights temporarily disrupt established downstream mappings. However, within three fine-tuning epochs, this variance resolved, leading to improved convergence boundaries and a measurable gain in validation accuracy.



E. Point of Impact Qualitative Assessment

The Hough-based POI module performed reliably on well-illuminated, high-contrast images, producing geometrically accurate impact origin estimates. The compiled 4-panel diagnostic overlay provides an auditable, visually clear summary of the geometric analysis, making it highly suitable for inclusion in case files and investigative presentations. However, localization accuracy degraded when processing images with heavy surface dust, significant blood patterns, or poor edge contrast. This degradation stems from the Canny operator's sensitivity to localized noise, which can introduce erratic edge artifacts and cause the Hough transform to map false line coordinates.



VI. LIMITATIONS AND FUTURE WORK

A. Current System Limitations

Despite its documented performance profiles, FractureVision v1.0 operates under several structural and environmental limitations that must be addressed prior to real-world deployment:

Dataset Scale Constraints:

The relatively small training collection (~200 images) limits the statistical robustness of the evaluation metrics and prevents the model from mapping the full variance of glass fracture morphology. Sensitivity to High-Velocity Variations:

The High-Velocity recall rate of 69.2% underscores a vulnerability to misclassifying atypical projectile patterns as low-velocity impacts.

Two-Dimensional Operational Scope:

The computer vision pipeline operates entirely on single flat photographs, ignoring valuable 3D topographic cues such as microscopic stress marks, concentric step fractures, or multi-angle depth indicators.

Dependence on Image Quality:

The deterministic POI module relies heavily on consistent lighting and clean fracture line extraction, making it sensitive to environmental noise and complex crime scene conditions.

B. Recommended Future Work

To overcome these current limitations, upcoming development cycles for FractureVision architecture will focus on the following research tracks: 1. 2. 3. 4. 5.

Dataset Expansion:

Partner with certified forensic crime laboratories and law enforcement academies to acquire a broader range of real-world ballistic and blunt-force glass fracture photographs.

Model Explainability:

Integrate Explainable AI frameworks, such as Grad-CAM gradient activations, to map the model's visual attention. This will allow expert examiners to visually verify which fracture characteristics drove the classification decision.

Uncertainty Quantification:

Implement calibrated probabilistic estimation techniques, such as Monte Carlo Dropout distributions, to output reliable confidence intervals alongside each classification.

Multi-Class Expansion:

Broaden the classification task into a multi-tiered framework that estimates projectile calibers and gauges impact angles based on entry-hole geometry and asymmetric fracture densities.

Mobile Field Deployment:

Pack the preprocessing, classification, and POI modules into a lightweight mobile application, enabling field officers to run immediate, preliminary assessments directly at crime scenes

VII. TOOLS AND TECHNOLOGIES

Table IV summarizes the software tools and libraries used in the development and evaluation of FractureVision.

TABLE IV
Software Tools and Libraries

Tool / Library	Purpose
Python 3.x	Core programming language
TensorFlow / Keras	Deep learning framework and model training
OpenCV (cv2)	Image preprocessing and Hough line transform
scikit-learn	Metrics: confusion matrix, AUC-ROC, classification report
NumPy	Numerical computation and array manipulation
Matplotlib / Seaborn	Visualization and result plotting
Google Colab (T4 GPU)	Cloud-based training environment

VIII. CONCLUSION

Cloud hardware environment utilized to accelerate network training and model tuning. FractureVision v1.0 demonstrates that automated deep learning models can classify glass fracture patterns and estimate spatial points of impact using compact, specialized datasets. By pairing a fine-tuned ResNet-50 backbone with a multi-phase transfer learning strategy, the system achieved a blind test accuracy of 80.8% and an AUC-ROC of 0.9231. Crucially, the model achieved a Low-Velocity recall rate of 92.3%, meeting a key forensic requirement by minimizing false projectile claims that could disrupt ongoing investigations.

Additionally, the deterministic Point of Impact module supplements binary classification with clear, geometric analysis, providing investigators with actionable spatial references for scene reconstruction. FractureVision is designed strictly as a decision-support system. It is intended to complement, rather than replace, human expertise by providing forensic examiners with a fast, objective, and reproducible first-pass assessment that can be easily reviewed and integrated into formal investigative reports

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